

Amorphous mesoporous calcium carbonate and magnesium carbonate as effective sorbents for the removal of phosphate in aqueous solutions

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Abstract

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Keywords

Phosphate removal, calcium carbonate, magnesium carbonate, adsorption

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Abstract

In this work, highly porous amorphous calcium carbonate (HPACC) and mesoporous magnesium carbonate (MMC) were tested as potential phosphate (PO_4^{3-}) sorbents in water. The performance of these sorbents at a PO_4^{3-} initial concentration between 0 – 1000 mg/L was evaluated. These highly porous materials were found to have enhanced PO_4^{3-} uptake at low concentrations (<100 mg/L) when compared with commercial CaCO_3 and MgCO_3 . The enhanced uptake on HPACC and MMC at low concentration was due to the high surface area and the porosity of these sorbents. The presence of NaCl salt of up to 1000 mg/L had very little effect on the performance of HPACC (<10% decreased uptake capacity), but the PO_4^{3-} uptake on MMC reduced to close to zero. HPACC with its high PO_4^{3-} uptake at low concentration could be relevant for real life application of PO_4^{3-} ions removal from water.

Keywords: Phosphate, Adsorption, Calcium carbonate, Magnesium carbonate

Introduction

Eutrophication of surface and freshwater sources remains a serious issue in today's society with wide reaching economic and environmental consequences. The weathering and erosion of bedrock represent natural processes that affect the phosphorus levels in many lakes [1]. However, it is undeniable that anthropogenic sources, such as agricultural run-off, municipal wastewater and industrial effluence, greatly contribute to the eutrophication of aquatic systems. The anthropogenic enrichment of phosphorus in such bodies of water not only contribute to the deterioration of the water quality but also lead to oxygen depletion due to the excessive growth of aquatic plants and microalgae. Resulting in detrimental effects on the marine ecosystem [2, 3]. As such, several methods have been developed for the removal of phosphate in aqueous systems, which can be divided into three principal classes, namely, biological and chemical processes and physical filtration. The use of biological processes mainly involves the removal of phosphate through plants and microorganisms, e.g. halophytes, polyphosphate accumulating organisms and microalgae. On the other hand, chemical processes rely on phosphate removal through precipitation and flocculation using metal salts and polymers. Finally, physical filtration typically utilizes membranes and sand. Some of these methods have been used with great success to treat elevated phosphate levels in water, e.g. the use of metal salts in wastewater treatment plants [4, 5]. However, despite their widespread use, many of these methods suffer from a number of drawbacks that limit their efficiency. These drawbacks include the production of heavy metal sludge and polluting by-products, narrow working conditions, limited removal efficiency, and high cost etc. [3, 6]. Adsorption processes offers an appealing alternative solution to some of these problems due to their high efficiency, low cost, minimal formation of polluting by-products, sorbent reusability and phosphate recoverability [6]. Phosphate

concentration below that of 0.1 mg/L may also be reached by using sorbents with high adsorption capacities[3]. Which could decrease phosphate-induced algal and cyanobacterial blooms and in turn prevent accelerated eutrophication. Various sorbents have been investigated in regards to phosphate adsorption in aqueous systems, ranging from biochars and activated carbons [7-9], metal-organic frameworks [10-12], metal oxides and hydroxides [13-18], to name a few. The application of calcium and magnesium carbonate-based sorbents for phosphate removal have however remained relatively unexplored despite their widespread abundance in nature. Studies have shown that phosphate ions are able to effectively interact with both Ca^{2+} -ions and Mg^{2+} -ions. In the case of calcite the interactions may results in either precipitation, forming calcium phosphate species, or adsorption which primarily takes place at concentrations below $0.2 \mu\text{mol PO}_4^{3-}/\text{g}$ [19, 20]. The uptake capacity of has been found to be limited by the number of specific adsorption sites on the accessible surface of the material. As the sites become occupied by adsorbed phosphate ions, lateral interactions between the adsorbed species and free ions occur, leading to the formation of clusters that may act as nucleation sites for the growth of calcium phosphate [20]. As such, factors restricting the use of calcite as an efficient sorbent for phosphate removal may be due to its limited adsorption capacity, particularly at high phosphate concentrations. The use of amorphous calcium and magnesium carbonates may offer an advantage compared to their crystalline counterparts in this regard due to their high surface areas. In recent years, our group has developed a number of highly porous inorganic materials with very high specific surface area. These materials include mesoporous magnesium carbonate (MMC) [21, 22] and highly porous amorphous calcium carbonate (HPACC) [23]. MMC and HPACC share a similar microstructure. They are constructed with nanometer-sized

particles that have aggregated together, forming a bulk particle with highly a porous structure. The high specific surface area and porosity allow these materials to have excellent sorption properties. Furthermore, the pore surface can be functionalized to tailor the chemistry of these materials. We have investigated and optimized these materials for a number of different applications including gas separation [24], drug delivery [23, 25-29], and adsorption of organic pollutants such as Azo dyes from water [30]. In this study, we will explore the utilization of MMC and HPACC as sorbents for the removal of phosphate from water. The high porosity of these materials, in conjunction with high affinity between calcium/magnesium ions and phosphate ions, could render them promising phosphate sorbents with high sorption kinetics.

2. Experimental section

2.1 Materials

Methanol (MeOH), Nitric acid (HNO₃, 65 %) and Calcium oxide (CaO) were purchased from VWR International AB, Sweden and Monopotassium phosphate (H₂KPO₄) and Magnesium oxide (MgO) were purchased from Sigma-Aldrich, USA.

All chemicals were used as received without further purification.

2.2 Synthesis of highly porous amorphous calcium carbonate (HPACC)

HPACC was prepared according to procedures previously published by our group [23]. In summary, 75 ml MeOH was placed into a glass reaction vessel (350 ml, Andrew glass co.) and heated in an oil bath at 50 °C while stirring under 4 bar of CO₂ pressure. Thereafter, 1.25 g of CaO was added to the warm MeOH and the reaction mixture was left stirring under the previously mentioned conditions for 4 hours. The obtained milky dispersion was thereafter centrifuged at 3,800 rpm for 15 minutes in order to remove

unreacted CaO particles, and a colloidal HPACC suspension was obtained. The HPACC suspension was then finally dried at 150 °C for 4 – 5 hours in a ventilated.

2.3 Synthesis of amorphous mesoporous magnesium carbonate (MMC)

MMC was synthesized according to procedures previously published by our group [21]. Briefly, 10 g of MgO was dispersed in 150 ml MeOH and the reaction mixture was left stirring in a glass reaction vessel (350 ml, Andrew glass co.) pressurized at 4 bar CO₂ for 24 h. The obtained cloudy mixture was centrifuged at 3,800 rpm for 30 minutes producing a yellow tinged colloidal MMC suspension. A pre-dried gel was then produced by heating the MMC suspension in a warm water bath under constant manual stirring, after which the gel was finally dried at 150 °C for 24 hours in a ventilated oven.

2.4 Surface area and porosity measurements

The specific surface area and porosity of the samples were determined by nitrogen sorption at 77 K using a Micromeritics ASAP 2020 Surface Area and Porosity Analyzer (Norcross, GA, USA). Samples were pre-treated at 373 K for 6 hours under dynamic vacuum (1×10^{-4} Pa) using a Micromeritics Smart VacPrep sample preparation unit (Norcross, GA, USA) prior to analysis.

2.5 Powder X-ray diffraction

Powder X-ray diffraction (PXRD) analysis was performed on a Bruker D8 Advanced TWIN/TWIN powder diffractometer (Billerica, WIS, USA) operated at 40 kV and 40 mA, using Cu α radiation ($\lambda = 1.5418$ Å), step-size 0.04° and a measuring time of 2 seconds per step.

2.6 Scanning electron microscopy

The morphology of the samples were studied using a Zeiss LEO 1530 scanning electron microscope (SEM) (Oberkochen, Germany) operated at kV. Samples were prior to imaging sputter-coated with a layer of Ag/Pd.

2.7 Inductive coupled plasma optical emission spectroscopy

Inductive coupled plasma optical emission spectroscopy (ICP-OES) analysis was carried out on a PerkinElmer Avio 200 ICP Optical Emission Spectrometer (Waltham, MA, USA) using normal argon gas flow and axial plasma. All standard and sample solutions were acidified to 2 v/v % using 65 % HNO₃ and filtered (0.45µm cellulose acetate membrane filters, VWR International AB, Sweden) prior to analysis. Blank solutions were acidified to 5 – 10 v/v % using 65 % HNO₃. Each analysis was carried out in triplicates and the measurements were done by element. Na- and P-standards were purchased from PerkinElmer.

2.8 Phosphate adsorption study

Phosphate adsorption studies were carried out on HPACC, MMC, commercial CaCO₃ and commercial MgCO₃ at room temperature. Each sorbent (1 mg/mL) was dispersed in a 15 ml or 50 ml Falcon tube containing phosphate solutions of concentrations between 0 – 1000 mg/ml. The dispersions were then left shaking on a Heidolph Multi Reax orbital shaker (Schwabach, Germany) at 1,000 rpm for 24 hours, after which they were filtered (0.45µm cellulose acetate membrane filters, VWR International AB, Sweden) and the phosphate concentration determined using ICP-OES.

The phosphate standard solution used in the study was prepared by dissolving approximately 1.43 g of KH₂PO₄ in 1000 ml ultra-pure water, resulting in final phosphate concentration of 1000 mg/ml. Phosphate solutions for the adsorption study

were prepared by diluting the standard solution by addition of ultra-pure water until the desired concentrations were obtained. The exact PO_4^{3-} concentration of each solution before adsorption experiments were determined by ICP-OES in order to obtain the correct PO_4^{3-} initial concentration of each individual experiments and to minimize the errors, as we had found that there was a large variation of PO_4^{3-} initial concentrations even between standard solutions prepared in the same way.

3. Results and discussion

3.1 Sorbent characterization

The synthesized HPACC and MMC were characterized in order to ensure the quality of the sorbents. The PXRD diffractograms displayed in Figure 1a show that HPACC and MMC were both X-ray amorphous and SEM images (Figure 2) show that the materials were constructed from aggregated spherical nanoparticles. The nitrogen sorption isotherms of HPACC and MMC (Figure 1b) both exhibit a reversible type IV shape indicative of mesoporous adsorbents according to IUPAC classifications [31]. The sorbents were found to be porous with a specific Brunauer–Emmett–Teller (BET) surface area of 343 and 561 m^2/g for HPACC and MMC, respectively. These observations are in good agreement with our previous findings, and for further discussions regarding the characterization of these materials, we refer the readers to our previous publications [21, 23].

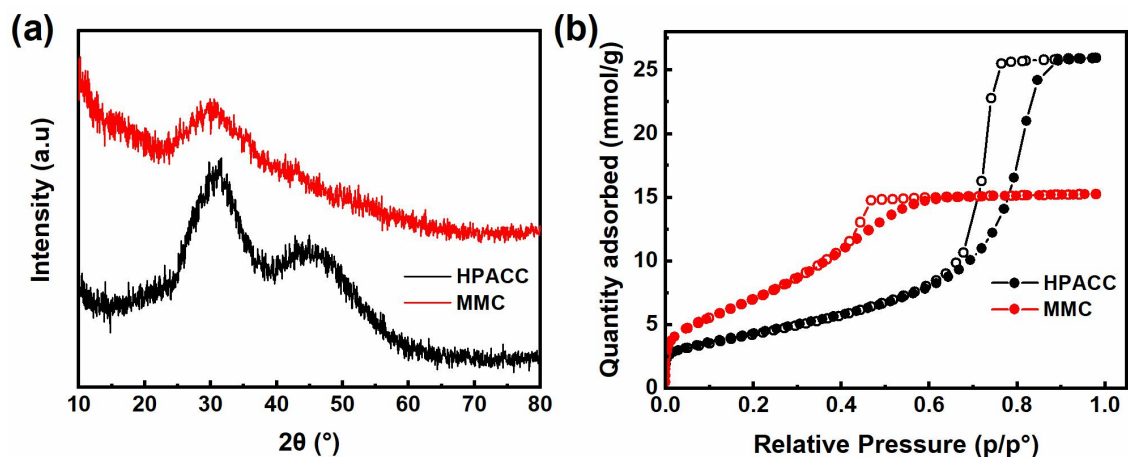


Figure 1. (a) PXRD diffractograms and (b) nitrogen sorption isotherms of HPACC and MMC.

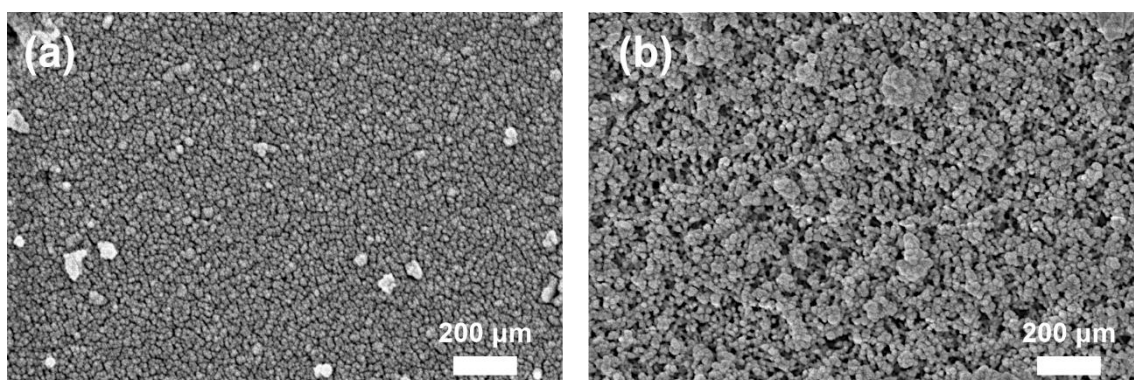


Figure 2. SEM images of (a) MMC and (b) HPACC.

3.2 Phosphate adsorption on HPACC and MMC

The phosphate uptake on HPACC, MMC, commercial CaCO_3 and commercial MgCO_3 at 24 h at initial concentrations of 10 – 1000 $\text{mg PO}_4^{3-}/\text{ml}$ are shown in Figure 3a-d. All tested sorbents displayed noticeable phosphate uptake. The trend in phosphate adsorption was found to be comparable for all materials, wherein the uptake of PO_4^{3-} increased with increasing PO_4^{3-} concentration.

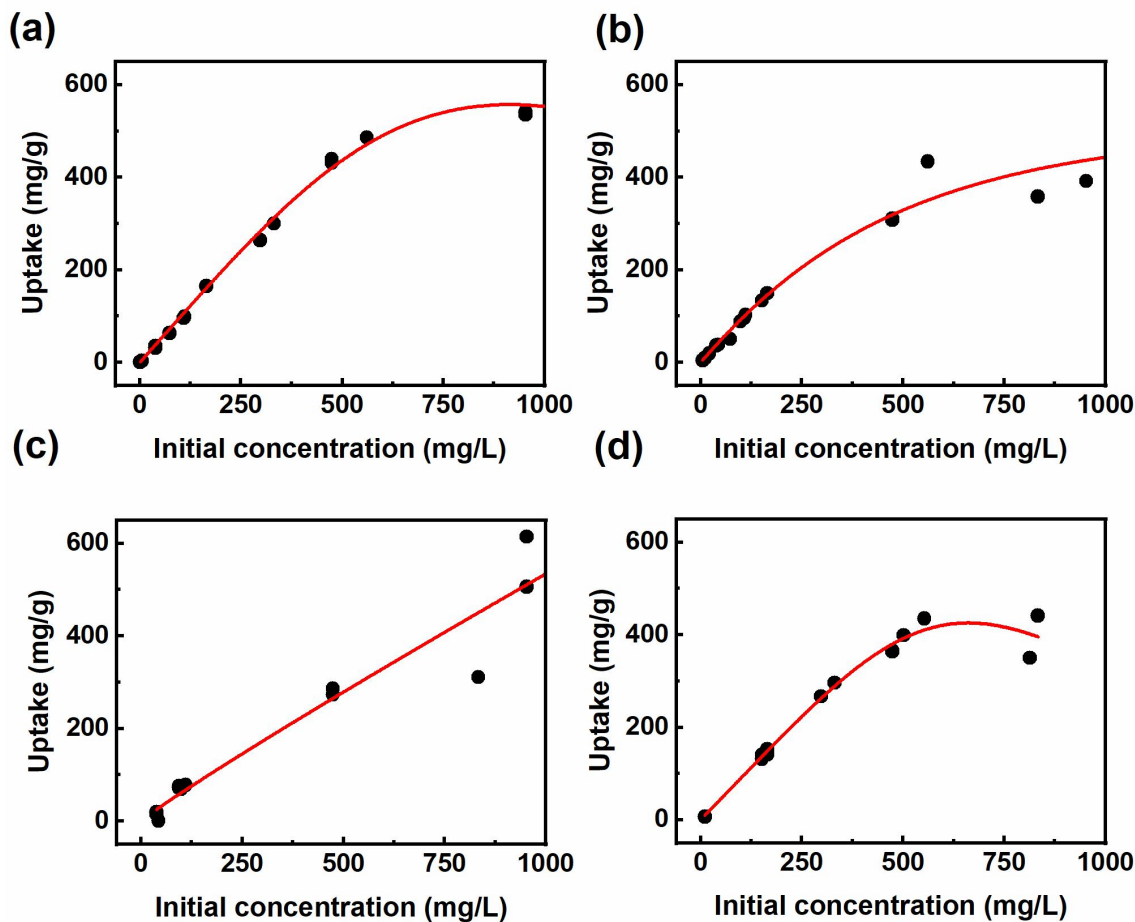


Figure 3. PO_4^{3-} uptake on a) HPACC, b) MMC, c) commercial CaCO_3 and d) commercial MgCO_3 varying initial PO_4^{3-} concentrations. The variation in the initial concentrations for the different samples were due to the difference between the expected and real concentrations in the standard solutions before the adsorption experiments, as discussed in the experimental section.

The equilibrium adsorption isotherms (Figure 4a-d) show comparable PO_4^{3-} uptakes for both the synthesized sorbents and their commercial counterparts at relatively high concentrations, more specifically above 300 mg PO_4^{3-} /L and 50 mg PO_4^{3-} /L for the CaCO_3 -based and MgCO_3 -based sorbents, respectively. An enhanced PO_4^{3-} adsorption was however observed on HPACC and MCC below these concentrations compared to commercial CaCO_3 and MgCO_3 . This observation could most likely be attributed to the

increased surface area of the HPACC and MMC sorbents when compared with CaCO_3 and MgCO_3 . The high surface area HPACC and MMC sorbents had an increased number of high-energy sorption sites within the pores of these sorbents. These high-energy sites would in turn lead to an enhanced surface interaction between the sorbents and PO_4^{3-} ions, particularly at low concentrations. Table 1 shows a comparison between HPACC, MMC and other sorbents. Table 1 compares the PO_4^{3-} uptake of the synthesized sorbents with those available in the literature. Unfortunately, given the difference in the initial concentrations used in the different study, it is difficult to make direct comparisons between the different sorbents. On the other hand, the sorbents presented in this study showed a reasonable level of PO_4^{3-} uptake both at low initial concentration and at high initial concentrations.

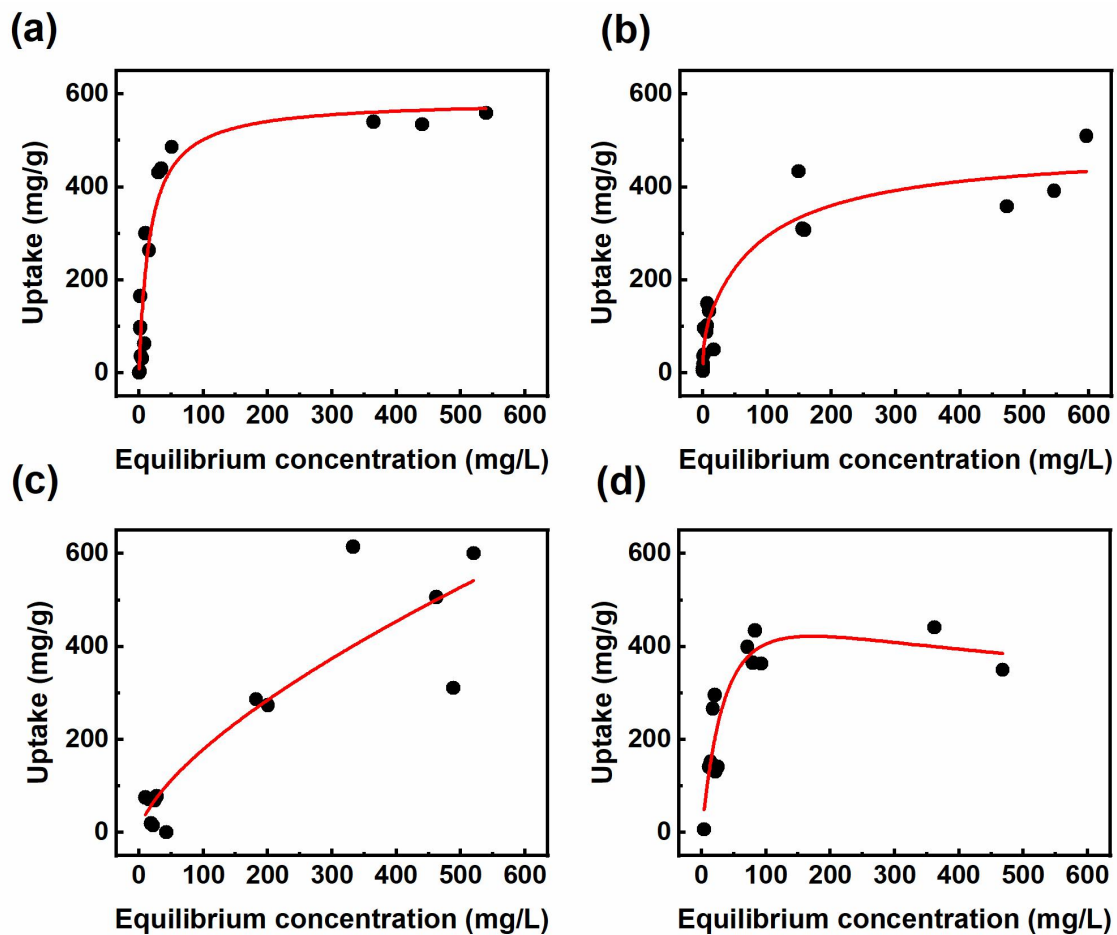


Figure 4. Equilibrium PO_4^{3-} uptake of a) HPACC, b) MMC, c) commercial CaCO_3 , and d) commercial MgCO_3 .

Table 1. Comparison between the phosphate uptake on the sorbents explored in the study and other materials.

Materials	Initial PO_4^{3-} concentration (mg/L)	Sorbent dosage (g/L)	PO_4^{3-} uptake (mg/g)	Reference
HPACC	953	1.1	540	This study
	38	1.0	36	
MMC	953	1.0	391	
	38	1.0	36	
Commercial CaCO_3	953	1.0	506	
	38	1.1	15	

Commercial MgCO ₃	833	1.0	440	
	38	1.0	7	
CP-Fe-I	50	2	22.1	
CTAB	10	2	1.2	
CTAB-IONP	10	2	4.9	[1]
nZVI	90	0.4	116	
Steel slag	11	40	0.2	
Fe-IRMOF-3-10	500	1	179	[32]
MIL-100-(Fe)	8	0.07	93.5	[33]
La@ZIF-8*	70	-	147.6	[34]
ZIF-8*	70	-	61.4	
Iron-impregnated WAS biochar*	1000	2	82.5	[35]
Fe-Mn binary oxide (Fe/Mn = 6:1)	10	0.2	26	[36]
1Fe-HNT	10	3	2.1	[37]
Lanthanum carbonate nanorods*	100	0.2	303	[38]
La(OH) ₃ /Fe ₃ O ₄ (4:1)*	15	0.1	74	[18]
LAH-1/10	80	1	68	[39]
Phoslock®*	10	1	10.6	[40]
Dolomite	100	2	48	[41]
am-ZrO ₂	10	0.1	64	[15]
Ti-OMS (20:1)	1	10	4.4	
Fe-OMS (20:1)	1	10	4.3	[42]
Zr-OMS (20:1)	1	10	4.4	
Al-OMS (20:1)	1	10	3.2	
Mg-ACC * (Mg/Ca = 4:6)	1425	5	198	[43]

3.3 Phosphate adsorption kinetics

We further tested the PO₄³⁻ uptake kinetics (with an initial PO₄³⁻ concentration of 100 mg/L) on these sorbents and the results are shown in Figure 5. Activated carbon was also added to this test for comparison. It is clear from Figure 5 that the increased

porosity on MMC and HPACC had a significant effect on the sorption rate and amount of PO_4^{3-} at this initial concentration level of 100 mg/L. HPACC showed the fastest relative uptake of PO_4^{3-} when compared with all the other tested sorbents. Over 80 mg/g of PO_4^{3-} uptake was recorded on HPACC within 2 hours, whereas all the other tested sorbents took up less than 10 mg/g within the same time. Although significantly lower than HPACC, MMC showed higher uptake than commercial MgCO_3 , CaCO_3 and activated carbon after 2 hours. After 16 hours the uptake of PO_4^{3-} on HPACC was around 100 mg/g, around 60 mg/g for MMC and around 40 mg/g for commercial CaCO_3 . The data presented here clearly shows that at low initial concentration (i.e. 100 mg/L), HPACC has significant advantages over the other tested sorbents both in terms of the uptake capacity and the uptake rate. This enhancement is likely to be the effect of the increased specific surface area and porosity.

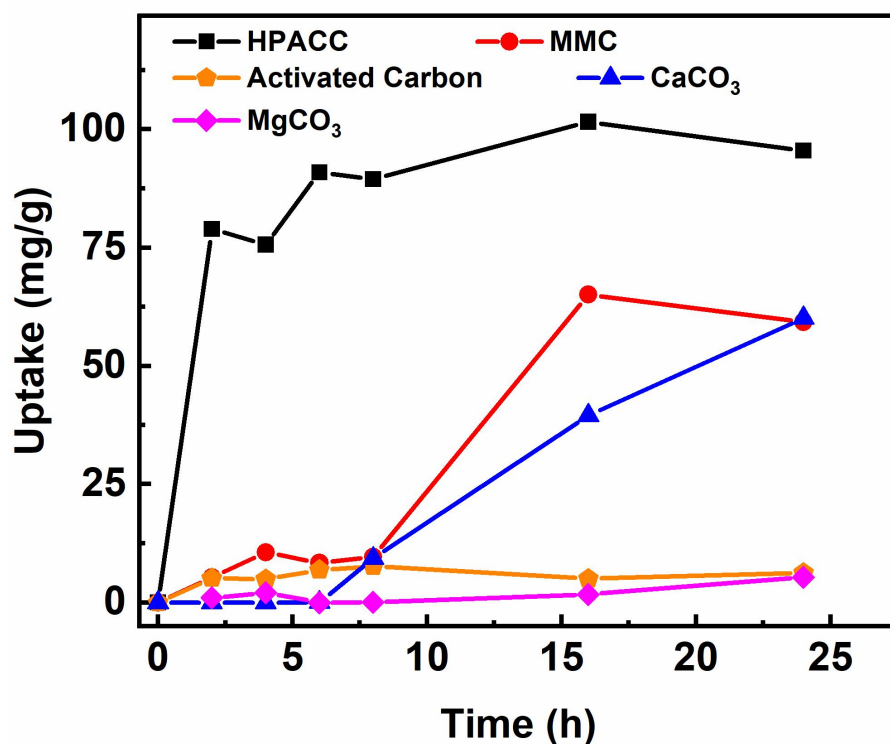


Figure 5. Phosphate adsorption kinetics of different sorbents. The initial PO_4^{3-} concentration was set at 100 mg/L.

3.4 The effect of salt

The effect of salt on the uptake of PO_4^{3-} on HPACC and MMC was tested by conducting the PO_4^{3-} sorption experiments on a 500 mg/L PO_4^{3-} solution with NaCl concentration of up to 1000 mg/L. Figure 6 shows that the uptake of PO_4^{3-} on HPACC was not significantly affected by the presence of NaCl, a slight decrease in the uptake of around 10% was noted when the NaCl concentration was over 50 mg/L. In contrast, the PO_4^{3-} uptake on MMC was noticeably affected by the presence of NaCl. The uptake of PO_4^{3-} on MMC decreased with increasing NaCl concentration and when the NaCl concentration reached 1000 mg/L, the PO_4^{3-} uptake was reduced to close to zero. These observations suggested that the Cl^- ions binds much more strongly to Mg than to Ca, as a result, the effect of NaCl on the PO_4^{3-} uptake on HPACC was not as significant.

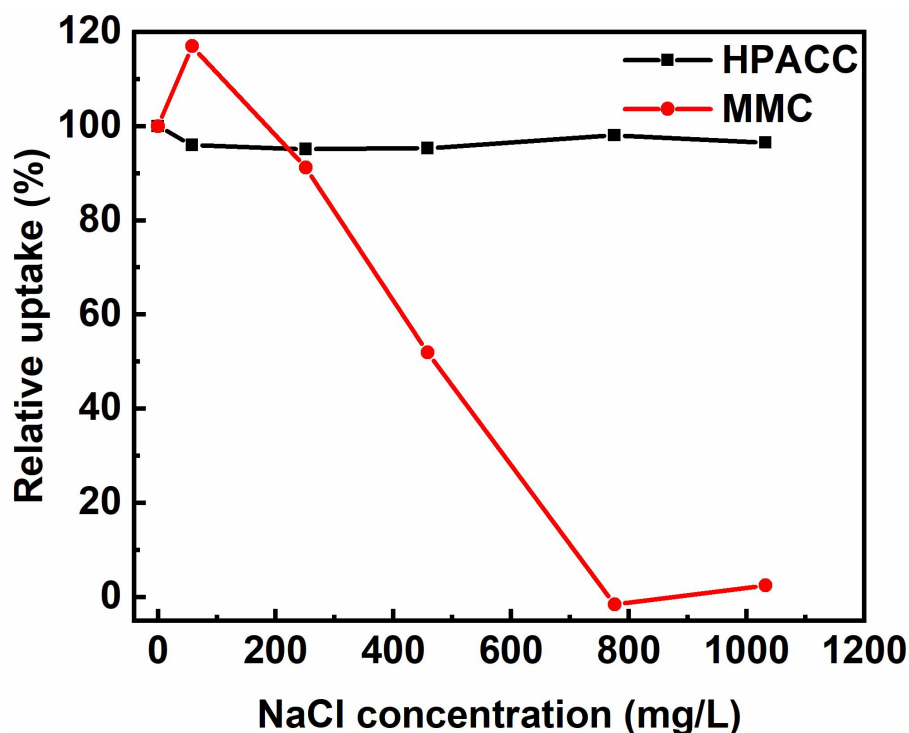


Figure 6. Phosphate adsorption kinetics of HPACC and MMC in the presence of NaCl salt, initial PO_4^{3-} concentration was set at 500 mg/L.

4. Conclusion

In this work, we tested the PO_4^{3-} ion sorption properties of a number of highly porous carbonate-based sorbents. Highly porous amorphous calcium carbonate, HPACC, showed very high uptake of PO_4^{3-} ions at low concentrations (< 100 mg/L). The increased porosity and surface area over the commercial CaCO_3 was responsible for the increased uptake at low PO_4^{3-} concentration. Mesoporous magnesium carbonate, MMC, also showed enhanced PO_4^{3-} uptake at low concentration over the commercial MgCO_3 . The presence of NaCl at up to 1000 mg/g had insignificant effects on the performance of HPACC, 90% of the PO_4^{3-} uptake was observed even at a NaCl concentration of 1000 mg/L. On the other hand, the PO_4^{3-} uptake capacity on MMC diminished when the NaCl concentration increased. HPACC could be further developed into an effective

PO₄³⁻ ion sorbent given its high porosity and specific surface area, which has shown enhanced PO₄³⁻ uptake over other sorbents, particularly at low PO₄³⁻ concentrations.

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Supporting information

Amorphous mesoporous calcium carbonate and magnesium carbonate as effective sorbents for the removal of phosphate in aqueous solutions

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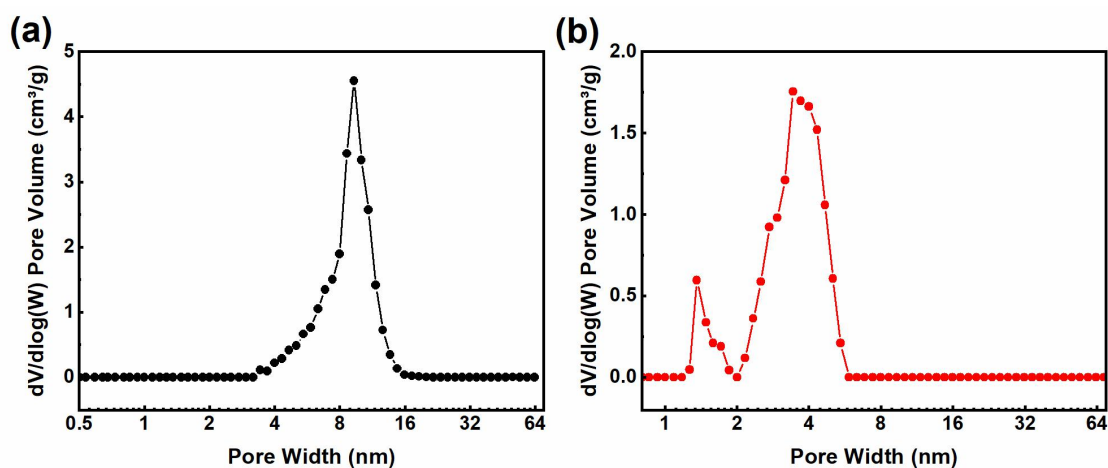


Figure S1. Pore size distributions of (a) HPACC and (b) MMC.

Table S1. Pore volume of HPACC and MMC.

Material	Total pore volume (cm³/g)
HPACC	0.868
MMC	0.482